

Multiple Choice

1. **Answer:** C

Options: A) $101.1 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; B) $26.75 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; C) $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$; D) $0.9377 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$

Note: The magnetogyric ratio is calculated by dividing the NMR frequency (198.52 MHz) by the magnetic field strength (750 MHz), resulting in a value of $7.081 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$.

2. **Answer:** D

Options: A) B; B) Al; C) In; D) Tl

Note: The +1 oxidation state is more stable for thallium (Tl) due to the inert pair effect, where the s-electrons are less likely to participate in bonding, leading to greater stability in the lower oxidation state compared to the higher +3 state.

3. **Answer:** A

Options: A) Neutrons; B) Alpha particles; C) Beta particles; D) X-rays

Note: Cobalt-60 is produced from cobalt-59 through neutron bombardment, where the absorption of a neutron transforms the stable isotope into the radioactive isotope used in cancer therapy.

4. **Answer:** D

Options: A) The process is exothermic.; B) The process does not involve any work.; C) The entropy of the system increases.; D) The total entropy of the system plus surroundings increases.

Note: The correct answer is based on the second law of thermodynamics, which states that for any spontaneous process, the total entropy of an isolated system, including both the system and its surroundings, must increase, indicating a natural tendency towards greater disorder.

5. **Answer:** D

Options: A) $6.345 \times 10^{-4} \text{ at } 0.335 \text{ T}$; $9.871 \times 10^{-5} \text{ at } 10.5 \text{ T}$; B) $0.793 \times 10^{-4} \text{ at } 0.335 \text{ T}$; $6.931 \times 10^{-7} \text{ at } 10.5 \text{ T}$; C)

Note: The correct answer reflects the calculation of proton polarization using the formula $P =$

$(B)/(kT)$, where B is the magnetic moment of the proton, B is the magnetic field strength, k is the Boltzmann

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because Zn_2^+ has a filled d-orbital configuration with no unpaired electrons, making it diamagnetic rather than paramagnetic.

7. **Answer:** True

Options: A) True; B) False

Note: Potassium hydrogen phthalate is considered a primary standard because it is stable, has a known and high purity, and can be dried to a constant weight, making it suitable for accurately determining the concentration of basic solutions in titrations.

8. **Answer:** True

Options: A) True; B) False

Note: The Larmor frequency of a proton in a magnetic field of 1 T is calculated using the formula $f = \gamma B$, where γ (the gyromagnetic ratio for protons) is approximately 42.58 MHz/T, confirming that at 1 T, the frequency is indeed 42.58 MHz.

9. **Answer:** True

Options: A) True; B) False

Note: The statement is true because magnesium oxide (MgO) has a high lattice enthalpy resulting from the strong electrostatic attraction between the doubly charged Mg^{2+} and O^{2-} ions, which leads to a greater charge density compared to other ionic compounds.

10. **Answer:** True

Options: A) True; B) False

Note: At standard temperature and pressure, bromine exists as a red-brown liquid due to its molecular structure and intermolecular forces, which allow it to remain in a liquid state while exhibiting volatility.

Long Form Response

11. **Answer:** The ^{13}C NMR spectrum of hexane exhibits three distinct chemical shifts due to the presence of different types of carbon environments within the molecule. Hexane, with the formula C_6H_{14} , has a linear structure where the carbon atoms are arranged in a chain. The three shifts correspond to the three types of carbon: the terminal methyl groups ($-\text{CH}_3$), the methylene groups ($-\text{CH}_2-$) adjacent to the methyl groups, and the central methylene groups that are surrounded by other methylene groups. The differences in chemical shifts arise from the electronic environments around each type of carbon, as the terminal carbons experience different shielding effects compared to the internal carbons. Thus, the molecular structure of hexane directly influences the distinct chemical shifts observed in its ^{13}C NMR spectrum.

Note: correct because it accurately identifies the distinct chemical environments of the carbon atoms in hexane—specifically the terminal methyl groups and the different types of methylene groups—leading to three unique chemical shifts in the ^{13}C NMR spectrum.

12. **Answer:** The ^{13}C NMR spectrum of 2-methylpentane displays five distinct chemical shifts due to the presence of five unique carbon environments within the molecule. The structure of 2-methylpentane includes a five-carbon straight chain (pentane) with a

methyl group (C) attached to the second carbon, leading to variations in the electronic environment of each carbon atom. Specifically, the terminal carbons (C_1 and C_5) are equivalent but distinct from the internal carbons (C_2 , C_3 , and C_4), each experiencing different degrees of shielding and deshielding effects. This unique arrangement results in five signals, corresponding to the different types of carbon atoms in the molecule, illustrating how molecular structure influences NMR spectra and emphasizing the role of isomerism in determining chemical shifts.

Note: correct because the five distinct chemical shifts in the ^{13}C NMR spectrum of 2-methylpentane correspond to the five unique carbon environments resulting from the molecule's structure, which includes a straight-chain pentane with a methyl substituent that creates variations in electronic environments among the carbon atoms.

End of Answer Key